# Section 12.1 - Vectors

MATH 230/231

## Agenda

- Welcome!
- Introductions, Syllabus & Expectations
- Section 12.1 - Vectors

## Planes and Vectors

A plane is a set of points $\{(x, y) : x, y \in \mathbb{R}\}$. The $xy$ coordinate system defines a plane. Then, a **vector**, $\mathbf{v}$ is determined by an initial point $P$ and a terminal point $Q$ and is denoted as follows:

$$\mathbf{v} = \overrightarrow{PQ}$$

Vectors are different from other mathematical objects because they possess both a **magnitude** and a **direction**.

For example, if I tell you I travel 30 mph for 3 hours, you will know I traveled 90 miles, but you will not know *where* I am. I would need to tell you the direction as well. This is a fundamental difference between speed and velocity. Velocity, often mistaken for speed, is a **vector** quantity.

# Parallel and Translated Vectors

Parallel Vectors:

Translation of a Vector:

## Components and Vector Length

Suppose that I have two points $P = (a_1, b_1)$ and $Q = (a_2, b_2)$. Then the vector $\overrightarrow{PQ}$ has components

$$\boxed{a = a_2 - a_1 \quad \text{(x-component)} \quad b = b_2 - b_1 \quad \text{(y-component)}}$$

We denote the pair of components as $\langle a, b \rangle$. For vector $\mathbf{v}$, we write $\mathbf{v} = \langle a, b \rangle$.

The **length of a vector** is defined as

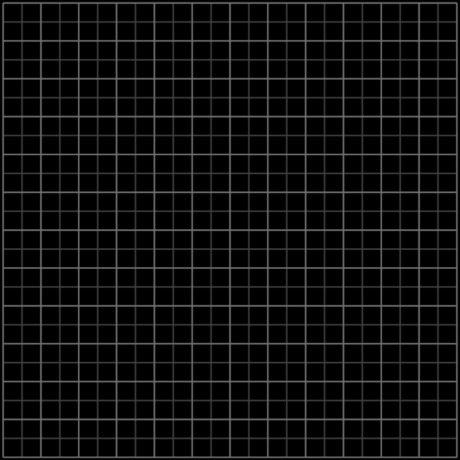
$$\boxed{||\mathbf{v}|| = ||\overrightarrow{PQ}|| = \sqrt{a^2 + b^2}}$$

Example

If $P = (1, 4)$ and $Q = (6, 2)$ , find the components of the vector and compute the length, or magnitude, of the vector.

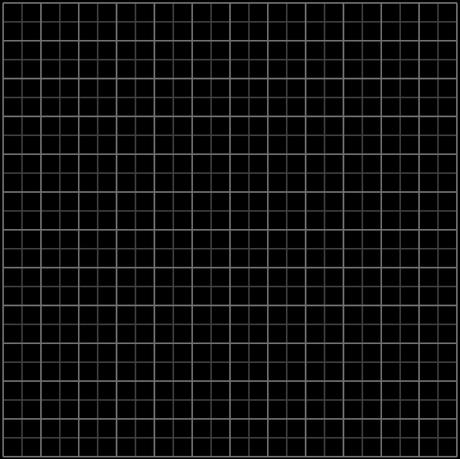
## Example

Sketch the vector on an $xy$-coordinate system.



## Visually Adding Vectors

The addition of two vectors **v** and **w** occurs according to the **Parallelogram Law**. Visually, we add vectors using the head-to-tail rule.



## Adding Vectors

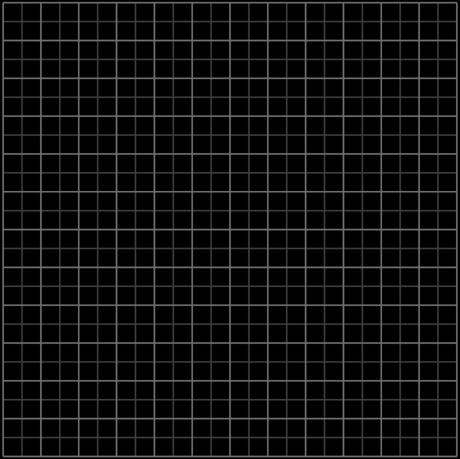
Adding two vectors is as simple as just adding the components. If $\mathbf{v} = \langle a, b \rangle$ and $\mathbf{w} = \langle c, d \rangle$ then,

$$\mathbf{v} + \mathbf{w} = \langle a + c, b + d \rangle$$

## Example

Add: $\mathbf{v} = \langle 1, 2 \rangle$ and $\mathbf{w} = \langle 5, 6 \rangle$.

- A vector $\mathbf{v}$ is **equivalent** to another vector $\mathbf{w}$ if $\mathbf{w}$ is a translation of $\mathbf{v}$.
- A vector $\mathbf{w}$ is **parallel** to $\mathbf{v}$ if and only if $\mathbf{w} = \lambda \mathbf{v}$ for some nonzero scalar $\lambda$.



## Vector Operations and Properties

**Operations:** If $\mathbf{v} = \langle a, b \rangle$ and $\mathbf{w} = \langle c, d \rangle$ then,

1. $\mathbf{v} + \mathbf{w} = \langle a + c, b + d \rangle$
2. $\mathbf{v} - \mathbf{w} = \langle a - c, b - d \rangle$
3. $\lambda \mathbf{v} = \langle \lambda a, \lambda b \rangle$
4. $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$

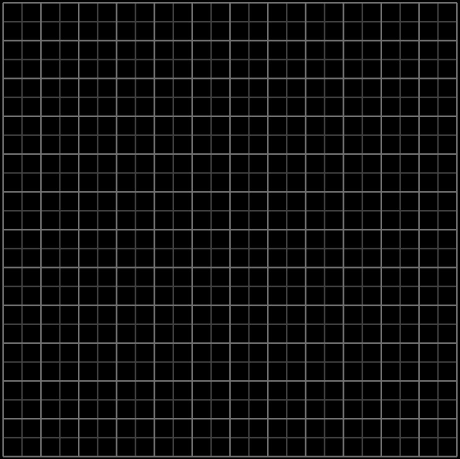
**Properties:** For all vectors **u, v, w** and for all scalars $\lambda$,

1. Commutative Law: $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$
2. Associative Law: $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
3. Distributive Law for Scalars: $\lambda(\mathbf{v} + \mathbf{w}) = \lambda \mathbf{v} + \lambda \mathbf{w}$

# Linear Combinations

A vector **u** is a linear combination of two other vectors **v** and **w** if there are scalar values $r$ and $s$ such that

$$\mathbf{u} = r\mathbf{v} + s\mathbf{w}$$



## Unit Vectors

For any vector $\mathbf{v}$ that is not the zero vector, the **unit vector** in the direction of $\mathbf{v}$ is defined as

$$\mathbf{e_v} = \frac{\mathbf{v}}{||\mathbf{v}||}$$

The **unit vectors in along the $x$ and $y$ axis** respectively are

$$\mathbf{i} = \langle 1, 0 \rangle \quad \& \quad \mathbf{j} = \langle 0, 1 \rangle$$

We can express a vector $\mathbf{v} = \langle a, b \rangle$ in standard basis notation as

$$\mathbf{v} = a\mathbf{i} + b\mathbf{j}$$

## Examples

Find the linear combination $\mathbf{u} = r\mathbf{v} + s\mathbf{w}$ of $\mathbf{v} = \langle 2, -5 \rangle$ and $\mathbf{w} = \langle 4, 0 \rangle$ if $r = 4$ and $s = -2$. Represent the vector in standard basis notation.

Find the unit vector $\mathbf{e}_{\mathbf{u}}$.

## Example

Verify that the unit vector $\mathbf{e_u}$ is, in fact, a unit vector.